JOHANSSON ENGINEERING

FLT Program — Regulatory Compliance Reference

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1. Purpose

This document is the **compliance proof** for the FLT program. It maps every program document (01-07) to the specific AFS 2017:3 sections it satisfies, including all three amendments. Use this document during audits, annual reviews, or when updating the program.

Regulatory basis: AFS 2017:3 consolidated with AFS 2019:1, AFS 2020:10, and AFS 2022:2.

2. Amendment Summary

All three amendments have been verified against the official PDF texts published by Arbetsmiljoverket.

AFS 2019:1 (in force 2 September 2019)

SECTION CHANGED	WHAT CHANGED	IMPACT ON FLT PROGRAM
1 Kap. §2	Scope clarified: vehicles excluded (EU directive refs added)	None — vehicles were never in scope
1 Kap. §6	Added definition: "enkla tryckkärl" (simple pressure vessels)	None — Doc 02 already classifies simple PVs
3 Kap. §1, §5, §7	CO2 fire extinguishers: max 10 years between inspections, specific int./ext. exam + pressure test	**Note:** If site has CO2 extinguishers, add to Doc 02 inventory
5 Kap. §3	First inspection must now include placement assessment (2 Kap §2-3)	**Covered:** Doc 02 section 6 includes placement risk
5 Kap. §4	When driftprov + int./ext. exam combined, driftprov done last (unless inspector says otherwise)	**Note:** Inform accredited body. No program doc change needed.
6 Kap. §11	Boiler monitoring: response time conditions changed from "1 AND 2" to "1 OR 2"	**Note:** Relevant only for boiler sites. Boilers meeting EITHER condition get extended response time.

AFS 2020:10 (in force 1 February 2021)

SECTION CHANGED	WHAT CHANGED	IMPACT ON FLT PROGRAM
Transitional point 4	Boiler operator certification: deadline extended to "5 years after entry into force"	**Superseded** by AFS 2022:2

AFS 2022:2 (in force 30 May 2022)

SECTION CHANGED	WHAT CHANGED	IMPACT ON FLT PROGRAM
Transitional point 4	Boiler operator certification deadline: **1 December 2025**	**Action:** All boiler operators in category 1-4 must be certified by 1 Dec 2025. Verify in Doc 01 / Doc 04 personnel section.

Key Transitional Provisions (from base AFS 2017:3, as amended)

POINT	REQUIREMENT	DEADLINE	STATUS
3	Lifetime journal (4 Kap §18) mandatory	1 Dec 2019	**Active** — Doc 06
4	Boiler operator certification (6 Kap §4)	1 Dec 2025	**Past deadline**
6	First inspection not required for pre-2017 equipment	Permanent	Noted
9	Equipment newly in scope must have first recurring inspection	1 Dec 2020	**Past deadline**

3. Program Document Compliance Map

Doc 01 — Coordination Responsibility

AFS SECTION	REQUIREMENT	HOW DOC 01 ADDRESSES IT
4 Kap. §14	Physical person must coordinate all work on Class A/B equipment	Section 2: named coordination person with contact details
4 Kap. §14	Written work instructions must exist	Section 4: responsibility checklist includes written instructions
4 Kap. §14	Must cover assembly, installation, maintenance, inspection	Section 4: all activities listed in responsibility checklist
4 Kap. §17	Physical person responsible for ensuring supervision is documented	Section 2: coordination person appointed and signed off

Doc 02 — Equipment Inventory

AFS SECTION	REQUIREMENT	HOW DOC 02 ADDRESSES IT
4 Kap. §15	Register of all Class A and B equipment at each workplace	Sections 1-3: vessel identification, technical data, classification
2 Kap. §6 pt 5	Equipment correctly marked	Section 7: documentation checklist includes nameplate
4 Kap. §10	Classification per PS x V and fluid group	Section 3: classification table with rules
4 Kap. §10	Air/nitrogen Class B exemption	Section 3: classification rules note the exemption
2 Kap. §2-3	Placement requirements	Section 6: placement risk assessment per vessel
AFS 2019:1 5 Kap. §3	First inspection includes placement assessment	Section 6: placement covered at inventory stage

Doc 03 — Risk Assessment

AFS SECTION	REQUIREMENT	HOW DOC 03 ADDRESSES IT
2 Kap. §1	Employer must assess risks considering 5 factors	Section 2: risk assessment per vessel
2 Kap. §1 pt 1	Experience from use	Covered in risk identification
2 Kap. §1 pt 2	Remaining lifetime information	Covered; links to Doc 06
2 Kap. §1 pt 3	Repairs and modifications	Covered in hazard identification
2 Kap. §1 pt 4	Accidents and incidents	Covered in hazard identification
2 Kap. §1 pt 5	Deviation reports and inspection results	Covered in hazard identification

Doc 04 — Supervision Program

AFS SECTION	REQUIREMENT	HOW DOC 04 ADDRESSES IT
4 Kap. §17	Documented supervision routines for Class A/B	The entire document IS the documented routine
4 Kap. §17	Must show compliance with 2 Kap §6 points 1-6	Section 1: all 6 mandatory checks listed with intervals
4 Kap. §17	Evaluate and revise at least once per year	Section 7: approval block with next review date
4 Kap. §17	Named physical person responsible	Section 5: personnel with named individuals
2 Kap. §6 pt 1	Equipment functions satisfactorily	Section 1, check 1
2 Kap. §6 pt 2	No leaks have occurred	Section 1, check 2
2 Kap. §6 pt 3	No harmful external or internal impact	Section 1, check 3
2 Kap. §6 pt 4	No other faults or deviations	Section 1, check 4
2 Kap. §6 pt 5	Correctly marked	Section 1, check 5
2 Kap. §6 pt 6	Prescribed inspections carried out	Section 1, check 6
4 Kap. §16	Continuous or periodic monitoring for Class A/B	Section 3: intervals defined per vessel and class
Sanction	Non-compliance: 10,000 — 100,000 SEK	Noted in document header

Doc 05 — Supervision Round Record

AFS SECTION	REQUIREMENT	HOW DOC 05 ADDRESSES IT
2 Kap. §6	Six mandatory checks must be performed	Section 2: checks 1-6 map directly to §6 points 1-6
4 Kap. §17	Supervision must be documented	The entire document IS the documentation
4 Kap. §19	Deviation reports for damage/deterioration	Section 3: findings with severity and action
4 Kap. §19 pt 1-7	Deviation report must contain 7 specific items	Section 3-4: description, severity, action, follow-up

Doc 06 — Lifetime Journal

AFS SECTION	REQUIREMENT	HOW DOC 06 ADDRESSES IT
4 Kap. §18	Journal showing remaining lifetime for Class A/B	Sections 3-7: design data, measurements, calculations, summary
4 Kap. §18	Different parts described separately if different lifetimes	Section 4: per-vessel measurement records with location detail
4 Kap. §18	Lifetime extension analysis required if design life reached	Section 7: lifetime summary with Safe/Monitor/Critical/Expired status
2 Kap. §1 pt 1-5	Previous experience assessment	Section 1: 5 risk factors assessed per vessel
2 Kap. §1 pt 2	Remaining lifetime information	Sections 5-7: corrosion rate, fatigue, creep tracking

Doc 07 — Annual Review

AFS SECTION	REQUIREMENT	HOW DOC 07 ADDRESSES IT
4 Kap. §17	Routines evaluated and revised at least once per year	The entire document IS the annual evaluation
4 Kap. §17	Must still show compliance with 2 Kap §6	Section 1: round completion tracking
4 Kap. §15	Equipment register kept current	Section 4: equipment changes tracked
AFS 2022:2	Boiler operator certification by 1 Dec 2025	Section 7: personnel competence status

4. Inspection Intervals (Bilaga 1)

Important: Inspection intervals are NOT fixed. The accredited inspection body determines intervals based on equipment condition. The table below shows the **base intervals** (Bilaga 1 §1.4.1). Actual intervals may be shorter or longer.

4.1 Driftprov (Operational Test) — Base Intervals

EQUIPMENT TYPE	BASE INTERVAL
Refrigeration/heat pump vessels	4 years
Air, nitrogen, noble gas vessels (Group 2a)	4 years
Vacuum-insulated condensed gas vessels	4 years
LPG storage vessels	4 years
Liquid-phase vessels (thermal expansion only)	4 years
All other pressure vessels	**2 years**
Cisterns	3 years
Piping connected to cisterns	3 years
Other piping	4 years

Extended: If safety equipment functioned without action at 2 previous tests, max interval = **4 years** (§1.4.2).

Shortened: If safety equipment did NOT function, next interval is **halved** (§1.4.3).

4.2 Internal/External Examination — Condition-Based

The inspection body sets the interval based on condition. Progressive system:

INTERVAL	KEY CONDITIONS
4 years	Not fire-affected, remaining life > 5 yr, low crack risk, non-harsh environment
6 years	Above + at least 1 clean 4-year inspection, no fatigue/creep
8 years	Above + at least 2 clean inspections, no corrosion/erosion
10 years	Above + at least 2 progressively longer inspections, no degradation
2 years	4-year conditions not met, but stable and > 3 yr before corrosion action
1 year	2-year conditions not met, but safe for 1 year per inspector
6 months	1-year conditions not met, but safe for 6 months

4.3 Exemptions from Internal/External Examination (5 Kap. §6)

EQUIPMENT	CONDITION
District heating heat exchangers	Temperature max 120 C
Vacuum-insulated CO2 vessels	—
Deep-cooled condensed gas vessels	Boiling point below -153 C
Hydraulic oil system vessels	Low crack risk material
Refrigeration/heat pump vessels	Non-corrosive medium
Water piping	Max 120 C, protected installation

5. Sanctions

VIOLATION	FINE
No documented supervision routines (4 Kap §17)	10,000 — 100,000 SEK
Pressurising without valid inspection (5 Kap §1)	15,000 — 300,000 SEK
Operating boiler without valid monitoring (6 Kap §1-2)	15,000 — 300,000 SEK

6. Annual Compliance Checklist

Use this checklist during the annual review (Doc 07) to verify the program remains compliant:

- [ ] **4 Kap §14:** Coordination person appointed and documented (Doc 01)
- [ ] **4 Kap §15:** Equipment register up to date (Doc 02)
- [ ] **4 Kap §16:** Monitoring type (continuous/periodic) documented per vessel (Doc 04)
- [ ] **4 Kap §17:** Supervision routines documented and reviewed this year (Doc 04, Doc 07)
- [ ] **4 Kap §17:** Named person responsible for supervision (Doc 01, Doc 04)
- [ ] **2 Kap §6:** All 6 mandatory checks addressed in every round (Doc 05)
- [ ] **4 Kap §18:** Lifetime journal maintained for limited-life equipment (Doc 06)
- [ ] **4 Kap §19:** Deviation reports completed for all damage/deterioration (Doc 05 findings)
- [ ] **2 Kap §1:** Risk assessment current and considers all 5 factors (Doc 03)
- [ ] **2 Kap §2-3:** Placement assessment completed for all vessels (Doc 02)
- [ ] **5 Kap §1:** All formal inspections current — no expired certificates
- [ ] **6 Kap §4:** Boiler operators certified (if applicable) — deadline was 1 Dec 2025
- [ ] **AFS 2019:1:** CO2 fire extinguishers included in inventory (if present on site)
- [ ] **AFS 2019:1:** First inspection includes placement assessment (5 Kap §3)

Revision History

REV	DATE	AUTHOR	DESCRIPTION
A	2026-04-11	N. Johansson	Initial release — verified against official PDF texts of AFS 2017:3, AFS 2019:1, AFS 2020:10, AFS 2022:2